

Basic Principles of Medicine 2

Module : Digestion and Metabolism

Lecture No: 40

Title: Clinical correlates and Inborn errors of metabolism - Flipped Classroom

Slides with answers will be posted after the interactive session

Please answer questions before the lecture

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Objectives: Clinical correlates and Inborn errors of Metabolism

SOM. MK.I.BPM2.3.DM.4.BCHM.1373	Differentiate between inborn errors of metabolism that cause hypoglycemia
SOM. MK.I.BPM2.3.DM.4.BCHM.1374	Predict clinical features and laboratory tests used for inherited Urea cycle disorders
SOM. MK.I.BPM2.3.DM.4.BCHM.1375	Discuss various clinical situations which result in increased anion gap metabolic acidosis and lactic acidosis.
SOM. MK.I.BPM2.3.DM.4.BCHM.1376	Differentiate between various disorders of muscle energy metabolism
SOM. MK.I.BPM2.3.DM.4.BCHM.1377	Review acid-base disorders associated with gastric outlet obstruction (pyloric stenosis), diarrhea, diabetic ketoacidosis, lactic acidosis, ethylene glycol and methanol poisoning.

Recommended reading

-  <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- [Clinical Cases | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)
 - Integrative cases: Case 2: Severe fasting hypoglycemia
 - Focused cases: Case 9: Elevated ammonia
- Korf textbook Genetics – Pg 188-189 (newborn screening)

Arterial blood gas analysis of a 25-year-old man show pH=7.2 (7.4); $\text{PCO}_2 = 25\text{mmHg}$ (40) and $\text{HCO}_3^- = 15\text{mEq/L}$ (24); Anion gap = 25. Which of the following disorders may result in these lab findings?

- | | |
|--|-----|
| A. Diabetic ketoacidosis | 17% |
| B. Lactic acidosis | 17% |
| C. Chronic obstructive pulmonary disease | 17% |
| D. Diarrhea | 17% |
| E. Pyloric stenosis | 17% |
| F. High altitude sickness | 17% |

Increased anion gap metabolic acidosis

- Calculation of anion gap: $\text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-)$; Normal: 8-14 mEq/L
- Increased anion gap metabolic acidosis
 - Ethylene glycol poisoning
 - Methanol poisoning
 - Lactic acidosis
 - Diabetic ketoacidosis
- Identify compounds accumulating
- Complications

Case report 1

- A 3-year-old child with severe diarrhea, has the following reports:
- Arterial blood gas analysis on admission
 - pH: 7.24 (7.35-7.45)
 - PCO_2 : 30 mmHg (33-45)
 - $[\text{HCO}_3^-]$: 16 mmol/L (22-28)
 - Na^+ : 135 mEq/L (136-146)
 - Cl^- : 108 mEq/L (95-105)
- Calculate anion gap:
- What is the acid base disturbance?
- Why is the serum bicarbonate reduced in this patient?
- Note that in **secretory diarrhea**: Serum Cl^- and K^+ levels may be low

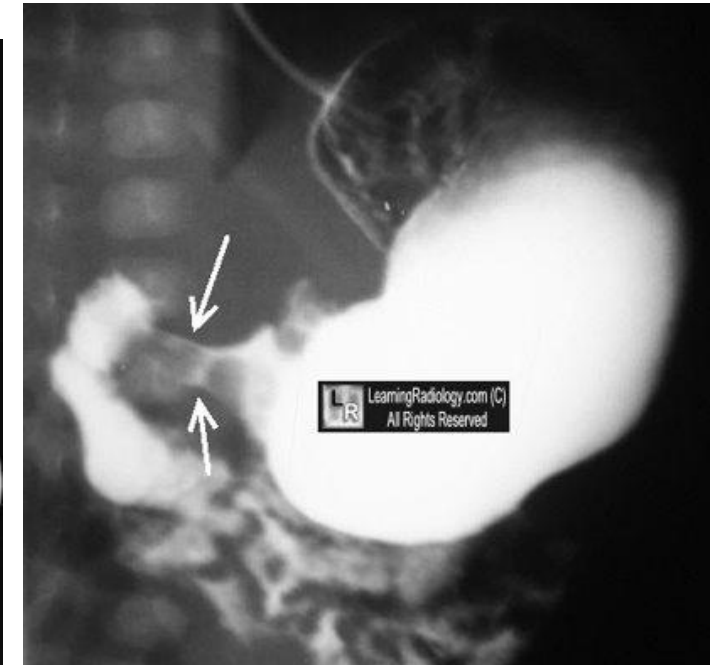
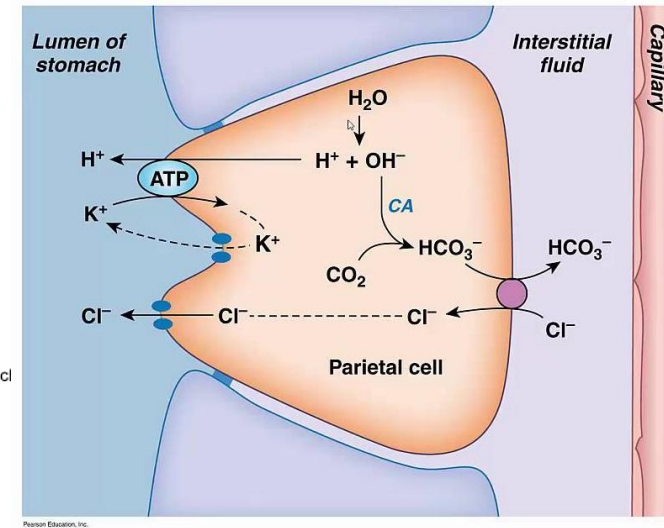
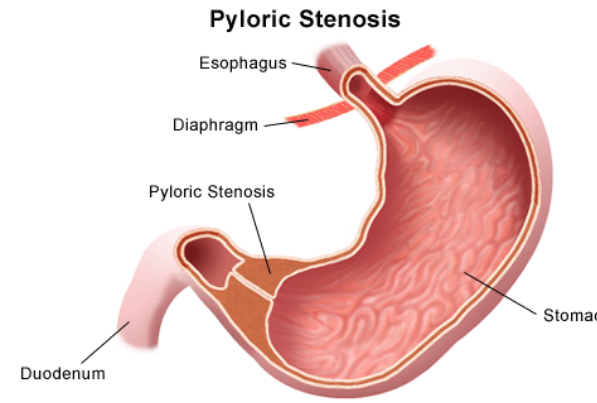
A 3-week-old boy is brought to the physician because of vomiting. Physical examination shows a mass in the epigastrium. Arterial blood gas reports: pH-7.5 (7.4); $\text{PCO}_2=48$ mmHg (40); $\text{HCO}_3^-=32$ mEq/L (24); $\text{Na}^+=130$; $\text{Cl}^-=95$. Barium swallow studies are shown. Which of the following disorders is most likely?

- | | |
|----------------------------------|-----|
| A. Pyloric stenosis | 17% |
| B. Lactic acidosis | 17% |
| C. Ketoacidosis | 17% |
| D. Diarrhea | 17% |
| E. Acute respiratory acidosis | 17% |
| F. Chronic respiratory alkalosis | 17% |

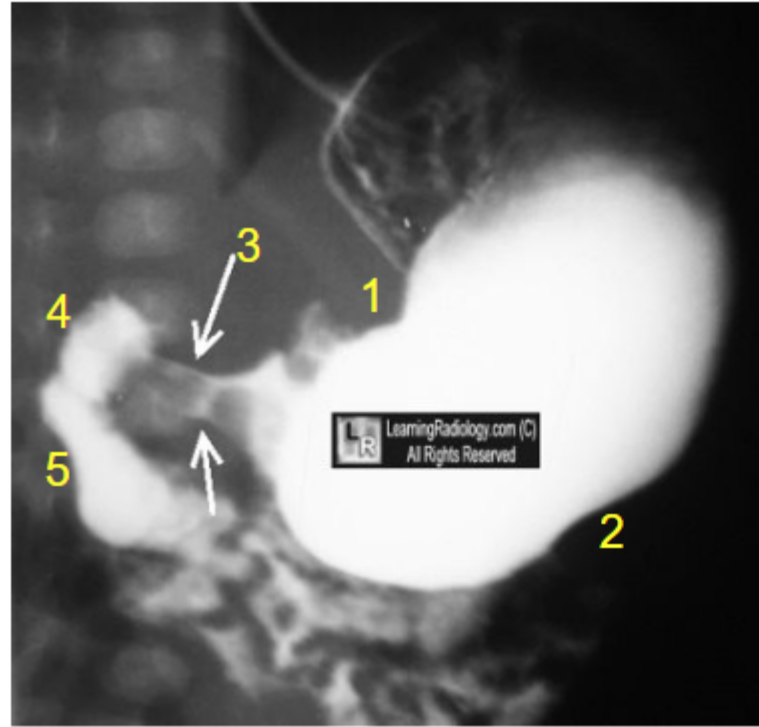


Case report 2

- A 3-week-old boy has been brought for consultation. He has had repeated episodes of **projectile** vomiting for the past two days.
- Arterial blood gas analysis on admission
 - pH: 7.56 (7.4)
 - PCO_2 : 50 mmHg (40)
 - $[\text{HCO}_3^-]$: 30 mmol/L (24)
- What is the acid base disturbance?
- Explain why there is an increase in arterial bicarbonate?
- Gastric outlet obstruction: Imaging



Congenital Hypertrophic Pyloric Stenosis



1. Lesser curvature
2. Greater curvature
3. Region of the pyloric sphincter
4. 1st part of duodenum
5. 2nd part of duodenum

Clinical presentation

- Non bilious **projectile** vomiting
- Starts ~3 – 6 weeks after birth (late as 5 months).
- After vomitus, infant is hungry and wants to feed again.

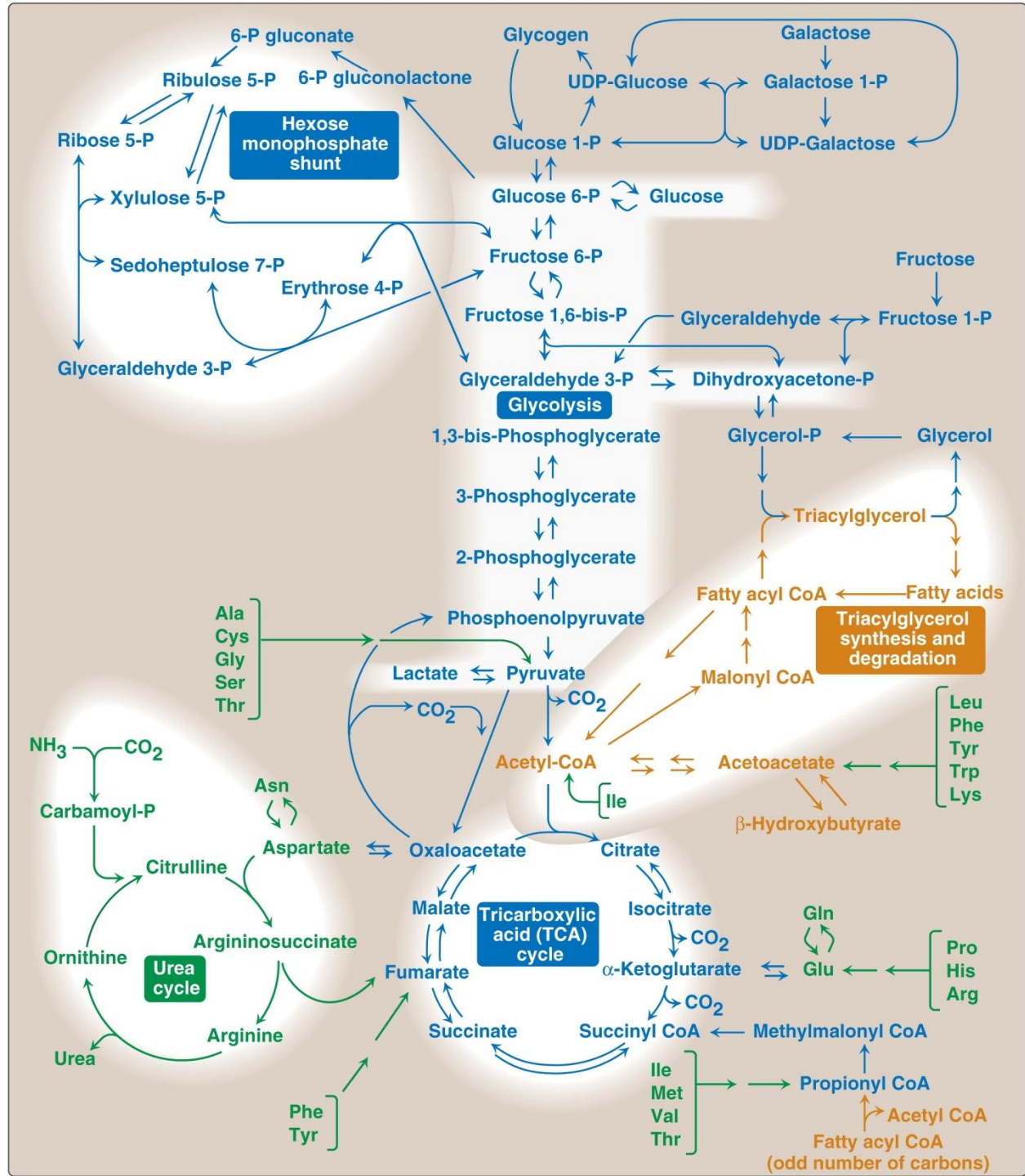
Physical examination

- Pyloric mass – olive shaped
- Visible gastric peristaltic wave

Imaging:

- **Barium swallow** – abnormal defect in filling in the region of the pyloric sphincter
- **Ultrasound** – thickening and lengthening of pyloric sphincter.

Intermediary metabolism

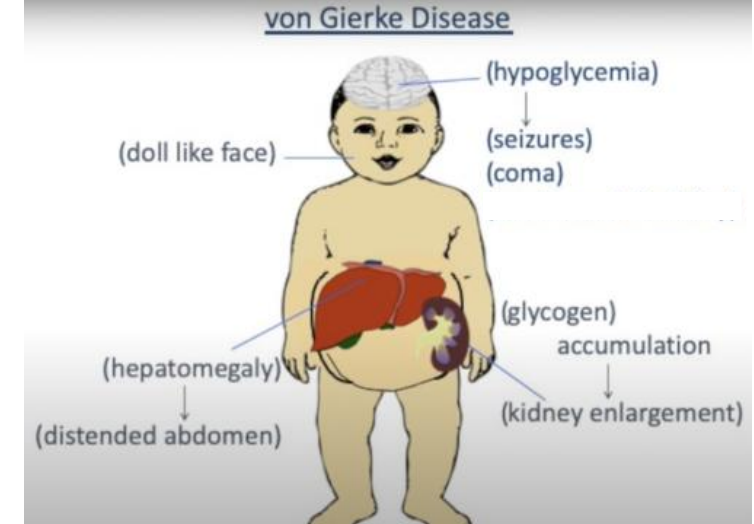


Case: 3 Severe fasting hypoglycemia

- JS is an 8-month-old infant who is brought to the physician because of twitching movements just before feeds in the morning.
- Physical examination: Appears sleepy and clammy to touch. Respiratory rate is increased. Temperature is normal. Weight is at 10th percentile.
- Lab tests on admission show blood glucose: 50 mg/dL (70-100 mg/dL)
- What inherited disorders are likely in this scenario?

Physical findings

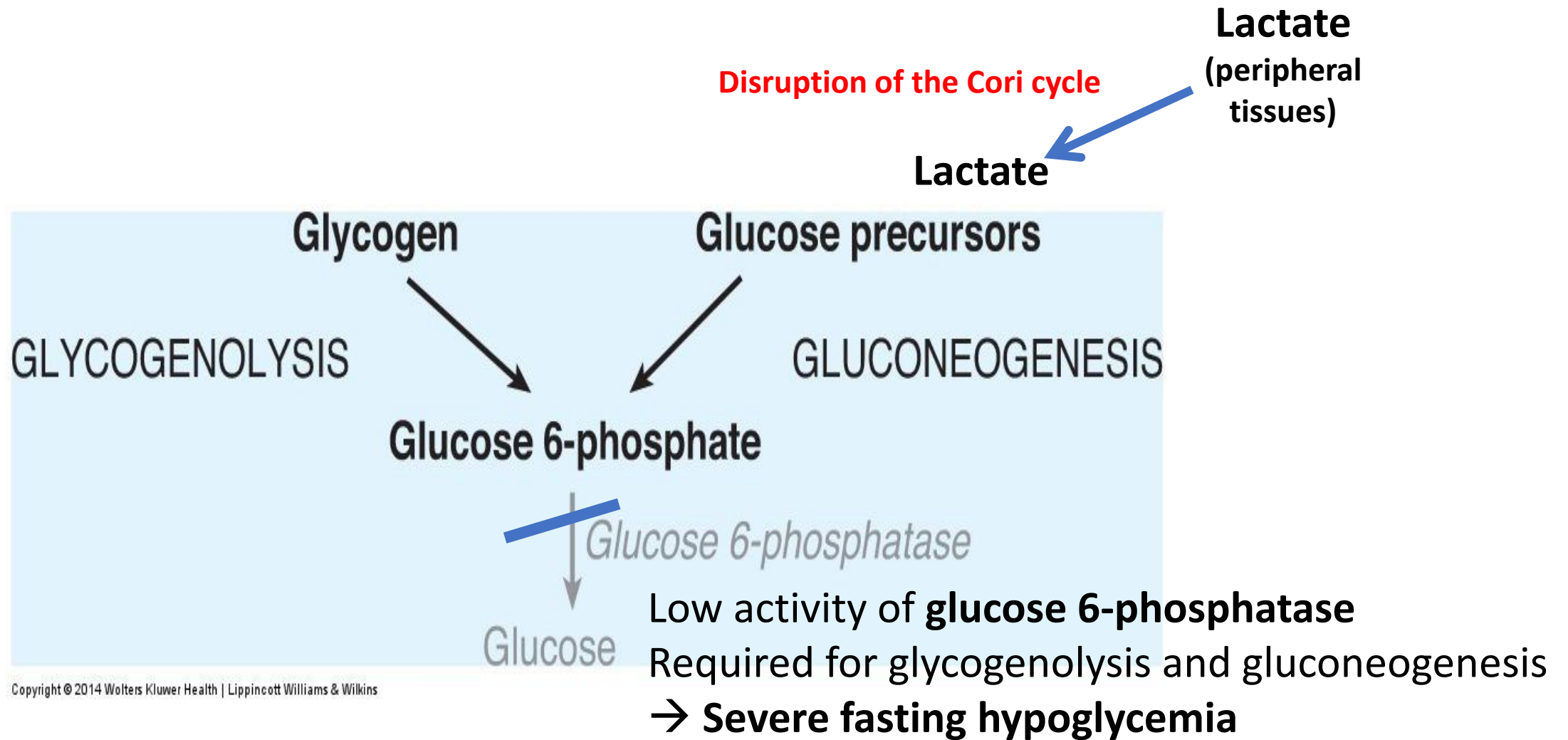
- Protuberant abdomen.
- Liver is palpable 4cm below right costal margin (hepatomegaly). Smooth and non-tender.
- Kidneys are enlarged and symmetrical
- Jaundice



Other lab tests:

- Lactate levels:
- Arterial pH= ; PCO_2 = and $[\text{HCO}_3^-]$ =
- Anion gap -
- Reducing sugar in urine:
- C8-C10 acylcarnitines:
- Dicarboxylic acids:
- Ketone body levels
- Liver enzymes

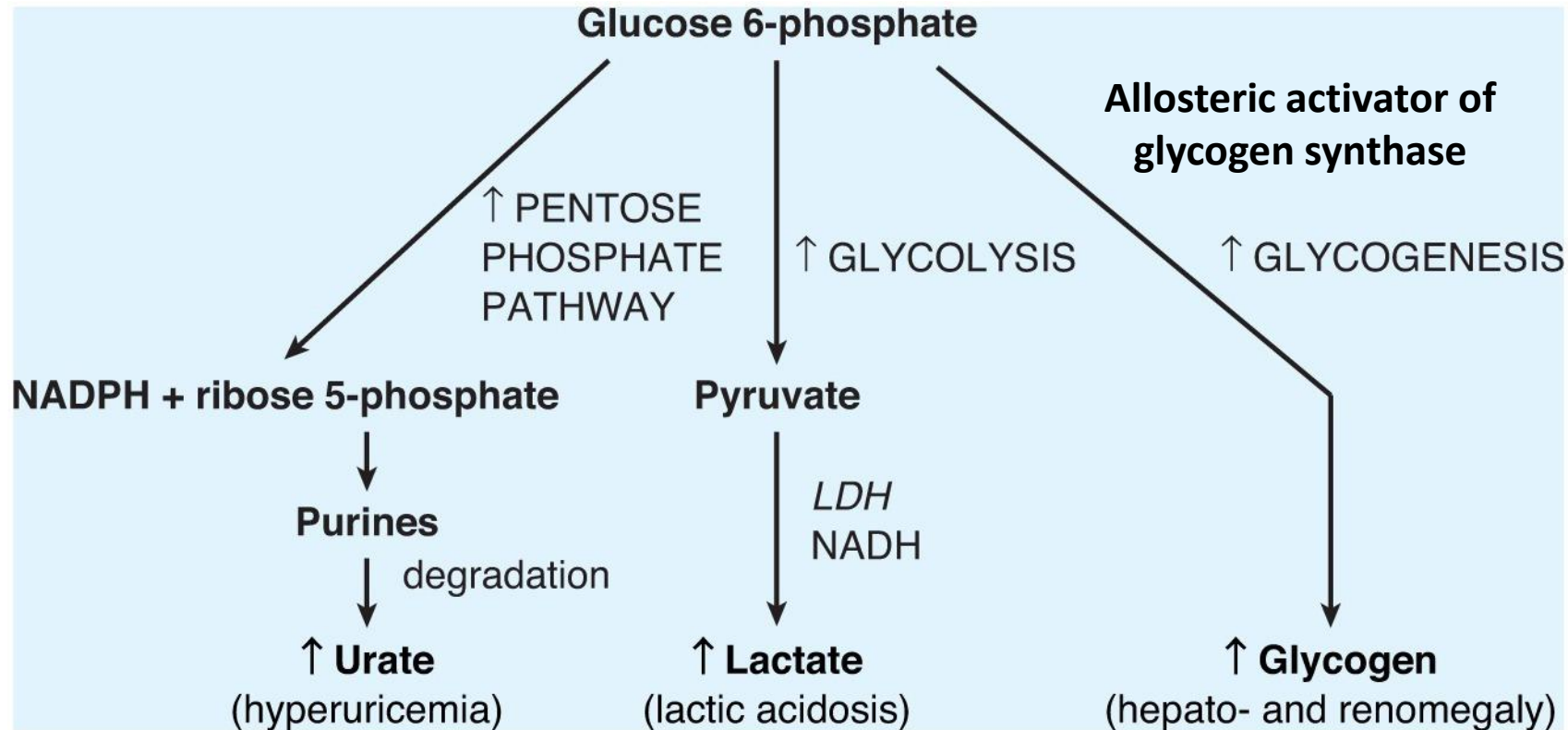
Von Gierke disease: Lactic acidosis



Von Gierke disease: Liver biopsy and enzyme studies

- Glucagon injection: 
- Liver Biopsy findings:
- Describe glycogen structure in:
 - Cori
 - Andersen
- Serum triglycerides: 280 mg/dL (Elevated)
- Serum Uric acid: elevated
- Effect of high Glucose 6-phosphate levels in hepatocytes
- Management:

Fates of accumulated glucose 6-phosphate in liver



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Differential diagnosis: **Hereditary fructose intolerance**

- Age of presentation:
- Enzyme deficient:
- Reducing sugar in urine:
- Metabolite accumulated in liver:
- Manifestations:
- Management:

Differential diagnosis: **Classical galactosemia**

- Age of presentation:
- Clinical signs:
- Reducing sugar in urine
- Toxic metabolite:
- Enzyme deficient:
- Management:

Differential diagnosis: **MCAD / CPT-I deficiency/ Primary carnitine deficiency (Systemic fatty acid oxidation defects)**

- Age of presentation
- Clinical features:
- Explain Hypoglycemia:
- Lab findings:

- **Management:**

A 6-month-old boy is admitted due to seizures. He had a respiratory infection three days earlier and has been refusing feeds for 2 days. On admission, blood glucose is 28mg/dL (70-100). Urine is negative for ketone bodies and positive for dicarboxylic acids. Which of following is the most likely diagnosis?

- | | |
|--|-----|
| A. Von Gierke disease | 17% |
| B. Pyruvate dehydrogenase complex deficiency | 17% |
| C. Systemic beta oxidation defect | 17% |
| D. CPT-II deficiency | 17% |
| E. McArdle disease | 17% |
| F. Classical Galactosemia | 17% |

Case: 4 Adolescent with muscle cramps

- 16-year-old boy has cramps on exercise. He says he passes a brown-colored urine for 2 days after strenuous activity (participating in sports)
- Lab tests – Blood glucose is normal
- Serum CK-MM activity is elevated
- Urine – brown color and contains myoglobin (rhabdomyolysis)
- What inherited disorders are likely in this scenario? Differential diagnosis

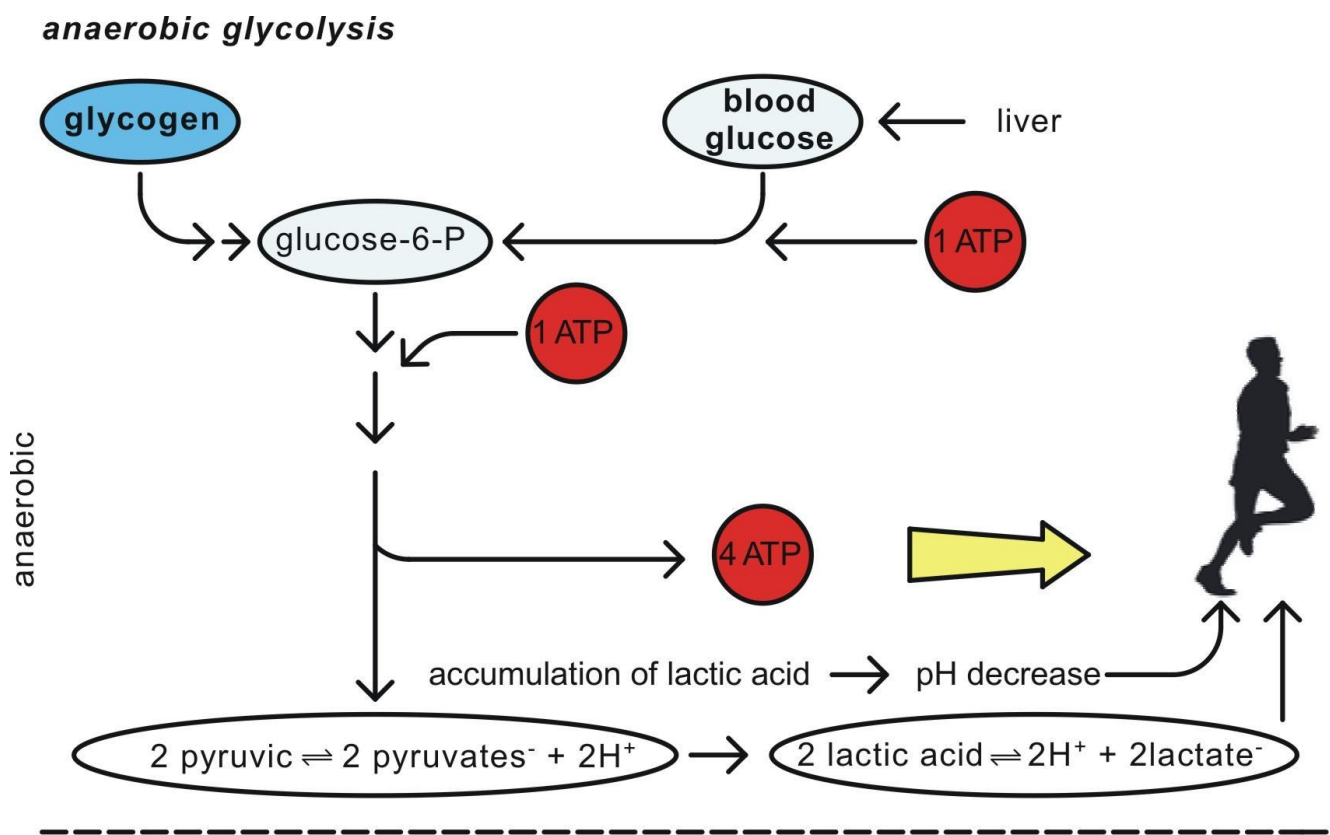


- What lab tests useful for differential diagnosis?



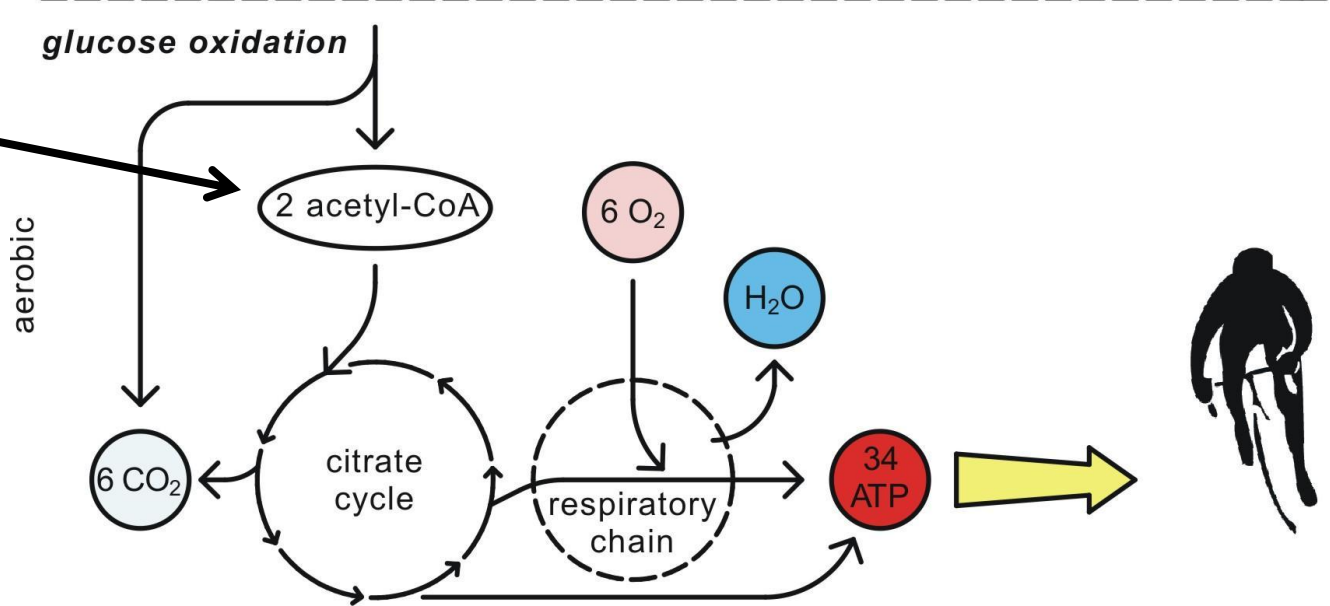
Metabolism in muscle: Aerobic and anaerobic

Glycogenolysis and
Glycolysis (Anaerobic)



Fatty acid oxidation

Aerobic: Fatty acid
oxidation and Krebs cycle



McArdle disease: Type V glycogen storage disease

- Forearm exercise test (Anaerobic)

- Muscle biopsy results:

- Enzyme deficient

- Which type of activity most affected? **Anaerobic/ Strenuous activity affected**

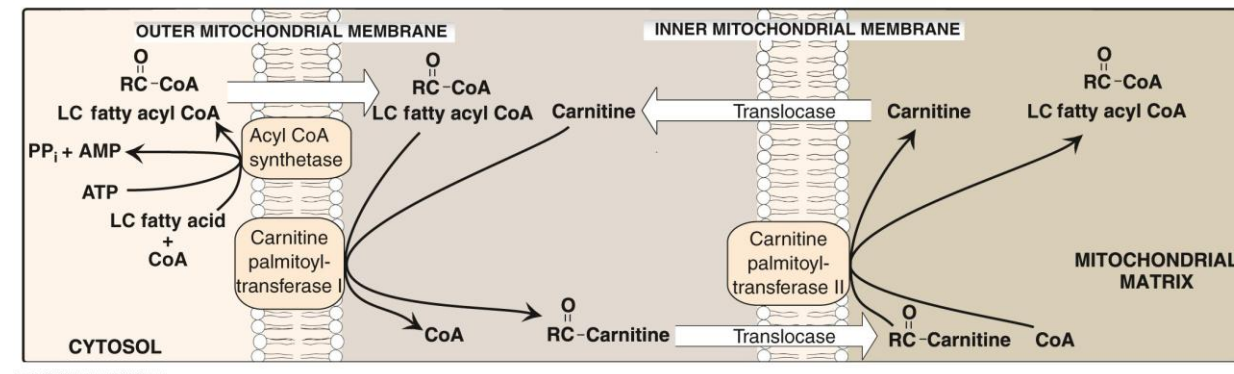
- Comment on Endurance (aerobic) muscle activity

- What is 'second wind phenomenon'?

Differential diagnosis: Myopathic fatty acid oxidation defect (CPT-II deficiency)

- Myopathic beta oxidation defect
- Muscle biopsy:
- Blood lactate levels following Forearm (Anaerobic) exercise test:

- Which type of activity most affected?
- Management: Carnitine supplements
- What's function of **carnitine**?



A 16-year-old boy complains of being excessively tired when he plays rugby. Forearm ischemic exercise test shows an increase in blood lactate levels. A muscle biopsy shows elevated cytosolic TAG levels esterified with long-chain fatty acids in the muscle. Which of the following disorders is most likely?

- | | |
|---|-----|
| A. Lipoprotein lipase deficiency | 17% |
| B. Familial hypercholesterolemia (Type IIA) | 17% |
| C. Refsum disease | 17% |
| D. Carnitine Palmitoyl Transferase-II (CPT-II) deficiency | 17% |
| E. McArdle disease | 17% |
| F. Zellweger syndrome | 17% |

An 8-month-old boy is brought to the emergency department because of inability to rouse. Physical examination shows lethargy and hypotonia. Laboratory studies are shown in the table. Which of the following gene mutations is most likely?

Parameter	Patients results (Normal)
pH	7.2 (7.4)
HCO ₃ ⁻	18 (24)
Blood glucose	75 (70-100)
Lactate	Increased

- A. Glucose 6-phosphatase 17%
- B. Pyruvate dehydrogenase complex 17%
- C. MCAD deficiency 17%
- D. Muscle glycogen phosphorylase 17%
- E. Galactose 1-P uridyl transferase (GALT) 17%
- F. Glucokinase 17%

A 2-year-old boy is brought to the physician because of a 6-month history of failure to thrive and a delay in the developmental milestones. Laboratory studies show elevated levels of an organic acid. Administration of thiamine results in clinical improvement. Deficiency of which of the following enzymes is most likely in this patient?

- | | |
|---|-----|
| A. Methylmalonyl CoA mutase | 20% |
| B. Branched chain keto acid dehydrogenase | 20% |
| C. Propionyl CoA carboxylase | 20% |
| D. Cystathionine synthase | 20% |
| E. Phenylalanine hydroxylase | 20% |

A newborn boy is brought to the NICU because of lethargy, difficulty feeding, and hyperventilation. He responds only to painful stimuli. Serum ammonia levels are elevated and orotic acid is detected in the urine. Which of the following intermediates would most likely be elevated in this patient? (What is the most likely enzyme deficiency?)

- | | |
|----------------------|-----|
| A. Homocysteine | 13% |
| B. Arginine | 13% |
| C. Glutamine | 13% |
| D. Alanine | 13% |
| E. N-acetylglutamate | 13% |
| F. Urea | 13% |
| G. Uric acid | 13% |
| H. Citrulline | 13% |



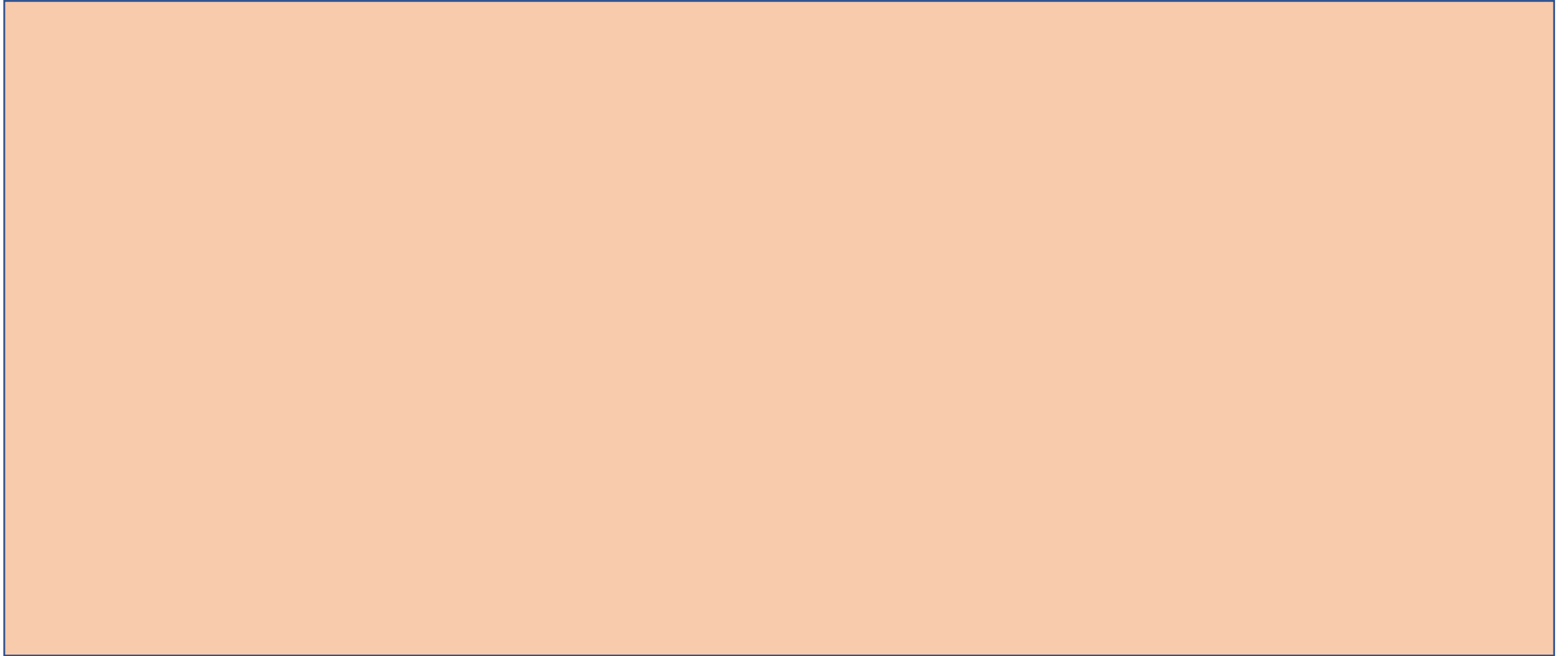
Case 5: Infant with lethargy

- A 2-day-old boy is irritable and lethargic.
- He has tachypnea and seizures
- Arterial blood gas: pH-high; PCO_2 – Low; HCO_3^- - Normal
- What is the acid-base abnormality?



- <http://emedicine.medscape.com/article/1174503-workup>
- Blood ammonia and glutamine levels are elevated
- What disorder is most likely?

Metabolic basis for neurotoxicity of hyperammonemia

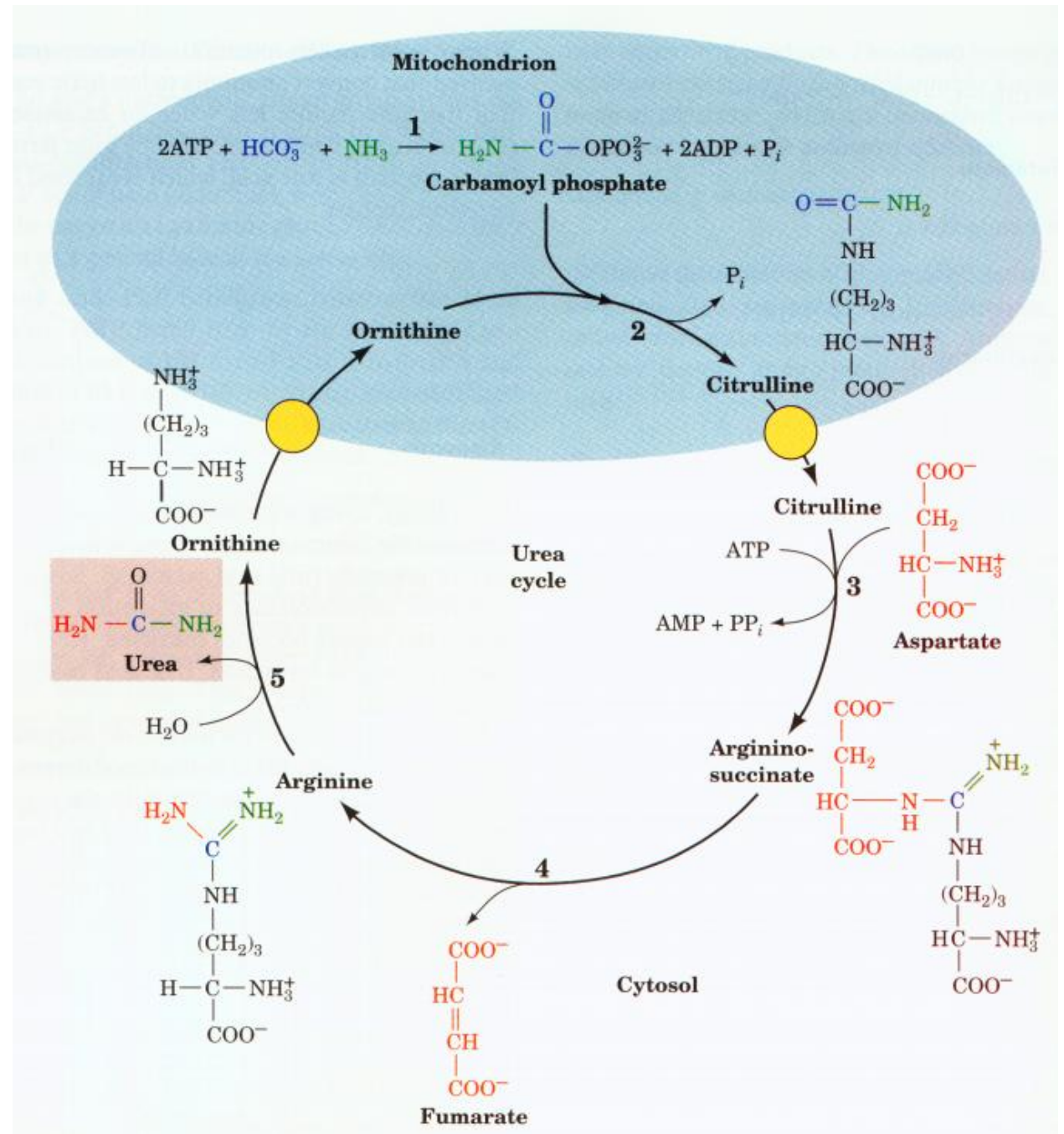


Urea cycle disorder: Serum Amino acid profile

- **Identify Urea cycle enzyme deficiency if:**
- Elevated glutamine (explain)
- Elevated argininosuccinate
- Elevated orotic acid (explain)
- Elevated citrulline
- Elevated arginine
- Low levels of all urea cycle intermediates (Elevated ammonia and glutamine)

Points to review

- Label all enzymes of the pathway shown
- Name activators of step 1
- Why is dietary protein restricted in children with UCD?
- Explain basis for Sodium benzoate/ Phenylbutyrate (phenylacetate) treatment



Orotic aciduria

- How do you differentiate between orotic aciduria due to urea cycle defect and due to pyrimidine biosynthesis defect based on
 - Clinical features
 - Lab findings



**THANK YOU FOR BEING
AMAZING STUDENTS**

AND

**BEST WISHES FOR
YOUR FUTURE**

And your EXAMS